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MOS TRANSISTOR AND METHOD OF MANUFACTURING THE SAME

Abstract

A method of manufacturing a MOS transistor includes: etching a surface of a SiC substrate with hydrogen gas in a state where the SiC substrate is heated to a temperature of 1200° C. or higher; forming an amorphous silicon layer on the surface of the SiC substrate after the etching; forming a gate insulating film made of silicon oxide on a surface of the amorphous silicon layer by a chemical vapor deposition; and performing a nitrogen termination treatment on the SiC substrate after the forming of the gate insulating film.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based on Japanese Patent Application No. 2024-032484 filed on Mar. 4, 2024, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a MOS transistor and a method of manufacturing the MOS transistor.

BACKGROUND

[0003] In a method for manufacturing a MOS transistor, a substrate made of SiC (silicon carbide) is etched with hydrogen gas (H.sub.2) in a silicon-rich atmosphere. As a result, the surface of the SiC substrate is cleaned, and a thin film of silicon crystal (Si) is formed on the surface of the SiC substrate. Next, a silicon oxide (SiO.sub.2) film is formed as a gate insulating film on the surface of the thin film of silicon. At this time, the thin film of silicon is oxidized, thereby suppressing oxidation of the surface of the SiC substrate. By suppressing the oxidation of the surface of the SiC substrate in this manner, the density of interface states at the interface between the silicon oxide film and the SiC substrate can be reduced. The density of interface states at the interface between the silicon oxide film and the SiC substrate is reduced, thereby improving channel mobility and decreasing channel resistance. This makes it possible to reduce the on-resistance of the MOS transistor.

SUMMARY

[0004] According to one aspect of the present disclosure, a method of manufacturing a MOS transistor includes: etching a surface of a SiC substrate with hydrogen gas in a state where the SiC substrate is heated to a temperature of 1200° C. or higher; forming an amorphous silicon layer on the surface of the SiC substrate after performing the etching with hydrogen gas; forming a gate insulating film made of silicon oxide on a surface of the amorphous silicon layer by chemical vapor deposition; and performing a nitrogen termination treatment on the SiC substrate after forming the gate insulating film.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0005] FIG. 1 is a cross-sectional view of a MOS transistor according to an embodiment.

[0006] FIG. 2A is a view for explaining a manufacturing method in the embodiment.

[0007] FIG. 2B is a view for explaining a manufacturing method in the embodiment.

[0008] FIG. 3A is a view for explaining a manufacturing method in the embodiment.

[0009] FIG. 3B is a view for explaining a manufacturing method in the embodiment.

[0010] FIG. 3C is a view for explaining a manufacturing method in the embodiment.

[0011] FIG. 3D is a view for explaining a manufacturing method in the embodiment.

[0012] FIG. 4 is an electron diffraction image of an amorphous layer.

[0013] FIG. 5 is a view showing a shoulder portion of a gate structure formed by a manufacturing method (high-temperature hydrogen etching for 15 minutes) in the embodiment.

[0014] FIG. 6 is a view showing a shoulder portion of a gate structure formed by a manufacturing method (high-temperature hydrogen etching for 30 minutes) in the embodiment.

[0015] FIG. 7 is a view showing a shoulder portion of a gate structure formed by a manufacturing method of a comparative example.

[0016] FIG. 8 shows results of evaluation in breakdown voltage relative to gate insulating films of FIGS. 5 to 7.

DETAILED DESCRIPTION

[0017] In a method for manufacturing a MOS transistor, a substrate made of SiC (silicon carbide) is etched with hydrogen gas (H.sub.2) in a silicon-rich atmosphere. As a result, the surface of the SiC substrate is cleaned, and a thin film of silicon crystal (Si) is formed on the surface of the SiC substrate. Next, a silicon oxide (SiO.sub.2) film is formed as a gate insulating film on the surface of the thin film of silicon. At this time, the thin film of silicon is oxidized, thereby suppressing oxidation of the surface of the SiC substrate. By suppressing the oxidation of the surface of the SiC substrate in this manner, the density of interface states at the interface between the silicon oxide film and the SiC substrate can be reduced. The density of interface states at the interface between the silicon oxide film and the SiC substrate is reduced, thereby improving channel mobility and decreasing channel resistance. This makes it possible to reduce the on-resistance of the MOS transistor.

[0018] After the silicon oxide film is formed, a nitrogen termination treatment is performed. The nitrogen termination treatment can further reduce the density of interface states at the interface between the silicon oxide film and the SiC substrate.

[0019] This specification proposes a method of manufacturing a MOS transistor with low on-resistance.

[0020] According to one aspect of the present disclosure, a method of manufacturing a MOS transistor includes: etching a surface of a SiC substrate with hydrogen gas in a state where the SiC substrate is heated to a temperature of 1200° C. or higher; forming an amorphous silicon layer on the surface of the SiC substrate after performing the etching with hydrogen gas; forming a gate insulating film made of silicon oxide on a surface of the amorphous silicon layer by chemical vapor deposition; and performing a nitrogen termination treatment on the SiC substrate after forming the gate insulating film.

[0021] In this manufacturing method, the surface of the SiC substrate is cleaned by etching with hydrogen gas. Thereafter, the amorphous silicon layer is formed on the surface of the SiC substrate. Next, the gate insulating film made of silicon oxide is formed on the surface of the amorphous silicon layer by chemical vapor deposition (hereinafter, referred to as CVD). When the gate insulating film is formed, the amorphous silicon layer is oxidized to form silicon oxide. This suppresses oxidation of the surface of the SiC substrate. Thereafter, the SiC crystal is terminated with nitrogen at the interface between the gate insulating film and the SiC substrate by nitrogen termination treatment. According to this manufacturing method, oxidation of the SiC substrate can be suppressed during the formation of the gate insulating film, so that the density of interface states at the interface between the gate insulating film and the SiC substrate can be reduced. Therefore, according to this manufacturing method, it is possible to reduce the channel resistance, and to manufacture a MOS transistor having a low on-resistance.

[0022] In an aspect of the present disclosure, a manufacturing method of a MOS transistor, a trench is provided on the main surface of the SiC substrate. In the step of etching the surface of the SiC substrate, a side surface of the trench is etched with hydrogen gas. In the step of forming an amorphous silicon layer, the amorphous silicon layer is formed on a side surface of the trench. In the step of forming the gate insulating film, the gate insulating film is formed on a surface of the amorphous silicon layer covering the side surface of the trench.

[0023] Accordingly, it is possible to suitably manufacture a trench-gate type MOS transistor having a low on-resistance, as will be described below. In a comparative hydrogen etching process, since a SiC substrate is etched with hydrogen in a silicon-rich atmosphere, a silicon thin film cannot be suitably formed on the side surface of the trench. In this trench-gate MOS transistor, it is not possible to reduce the density of interface states at the interface between the gate insulating film and the SiC substrate (that is, the side surface of the trench). With the technique in the comparative example, it is difficult to reduce the channel resistance in the trench-gate MOS transistor, and it is difficult to manufacture a trench-gate MOS transistor with a low on-resistance.

[0024] In contrast, according to the present disclosure, the side surface of the trench is cleaned by etching with hydrogen gas. Thereafter, an amorphous silicon layer is formed on the side surface of the trench. The amorphous silicon layer can be suitably formed on the sidewalls of the trench. Next, a gate insulating film made of silicon oxide is formed on the surface of the amorphous silicon layer by chemical vapor deposition (hereinafter, referred to as CVD). When the gate insulating film is formed, the amorphous silicon layer is oxidized to form silicon oxide. This restricts the side surfaces of the trench from being oxidized. Thereafter, the SiC crystal is terminated with nitrogen at the interface between the gate insulating film and the SiC substrate by nitrogen termination treatment. In this way, according to the manufacturing method of the present disclosure, oxidation of the side surface of the trench can be suppressed during formation of the gate insulating film, thereby reducing the density of interface states at the interface between the gate insulating film and the SiC substrate. Therefore, according to this manufacturing method, it is possible to reduce the channel resistance, and to manufacture a trench-gate MOS transistor having a low on-resistance.

[0025] For example, the etching with hydrogen gas etches the main surface of the SiC substrate in addition to the side surface of the trench. In the step of forming the amorphous silicon layer, the amorphous silicon layer is formed not only the side surface of the trench but also on the main surface of the SiC substrate. In the step of forming the gate insulating film, the gate insulating film is formed on the surface of the amorphous silicon layer in an area covering the side surface of the trench and the main surface of the SiC substrate.

[0026] Further, the amorphous silicon layer may be formed by chemical vapor deposition.

[0027] This configuration allows the amorphous silicon layer to be formed suitably.

[0028] For example, the amorphous silicon layer may be formed in a nitrogen atmosphere by chemical vapor deposition in a state where the SiC substrate is heated to a temperature of 400° C. or more and 550° C. or less.

[0029] This configuration allows the amorphous silicon layer to be formed suitably.

[0030] For example, the amorphous silicon layer may be formed by chemical vapor deposition in a state where the SiC substrate is placed under a pressure of 0.3 Torr or more and 1.0 Torr or less.

[0031] This configuration allows the amorphous silicon layer to be formed suitably.

[0032] For example, the amorphous silicon layer on the side surface of the trench may have a thickness of 6 nm or less.

[0033] If the amorphous silicon layer is too thick, a part of the amorphous silicon layer may remain unoxidized. In contrast, when the thickness of the amorphous silicon layer is 6 nm or less, it is possible to restrict the amorphous silicon layer from remaining without being oxidized.

[0034] For example, the main surface of the SiC substrate is a surface of an epitaxial layer.

[0035] Further, the nitrogen termination treatment may involve heating the SiC substrate to a temperature of 1200° C. or higher in nitrogen gas or nitrogen oxide gas.

[0036] For example, after formation of the gate insulating film, the main surface of the SiC substrate and the side surface of the trench are connected by an arc having a minimum radius of curvature of 53 nm or more in a cross section along a direction perpendicular to the trench in the main surface of the SiC substrate and along a thickness direction of the SiC substrate.

[0037] According to this configuration, current leakage at the boundary between the main surface of the SiC substrate and the side surface of the trench can be suppressed.

[0038] For example, after formation of the gate insulating film, the main surface of the SiC substrate and the side surface of the trench are connected by an elliptical or circular arc with an aspect ratio of 0.29 or less in a cross section along a direction perpendicular to the trench on the main surface of the SiC substrate and along a thickness direction of the SiC substrate.

[0039] According to this configuration, current leakage at the boundary between the main surface of the SiC substrate and the side surface of the trench can be suppressed.

[0040] According to another aspect of the present disclosure, a MOS transistor includes: a SiC substrate having a trench provided on a main surface thereof; a gate insulating film covering a side

surface of the trench; and a gate electrode provided in the trench. In a cross section along a direction perpendicular to the trench on the main surface and along a thickness direction of the SiC substrate, the main surface of the SiC substrate and the side surface of the trench are connected with each other by an arc having a minimum radius of curvature of 53 nm or more.

[0041] According to this configuration, current leakage at the boundary between the main surface of the SiC substrate and the side surface of the trench can be suppressed.

[0042] According to another aspect of the present disclosure, a MOS transistor includes: a SiC substrate having a trench provided on a main surface thereof; a gate insulating film covering a side surface of the trench; and a gate electrode provided in the trench. In a cross section along a direction perpendicular to the trench on the main surface and along a thickness direction of the SiC substrate, the main surface of the SiC substrate and the side surface of the trench are connected with each other by an elliptical or circular arc having an aspect ratio of 0.29 or less.

[0043] According to this configuration, current leakage at the boundary between the main surface of the SiC substrate and the side surface of the trench can be suppressed.

[0044] FIG. 1 shows a MOS transistor **10** manufactured by a manufacturing method according to an embodiment. The MOS transistor **10** includes a SiC substrate **12**, a gate insulating film **14**, a gate electrode **16**, an interlayer insulating film **18**, a source electrode **20**, and a drain electrode **22**. The SiC substrate **12** has an upper surface (main surface) **12a** and a lower surface **12b**. Trenches **24** are provided in the upper surface **12a** of the SiC substrate **12**. The trenches **24** extend linearly and parallel to each other on the upper surface **12a**. The gate insulating film **14** covers the inner surface of each trench **24** (specifically, the side surface **24a** and the bottom surface **24b** of the trench **24**). The gate electrode **16** is disposed in each trench **24**. The gate electrode **16** is insulated from the SiC substrate **12** by the gate insulating film **14**. The interlayer insulating film **18** covers an upper surface of the gate electrode **16**. The source electrode **20** covers the upper surface **12a** of the SiC substrate **12** and the upper surface of the interlayer insulating film **18**. The source electrode **20** is insulated from the gate electrode **16** by the interlayer insulating film **18**. The drain electrode **22** covers the lower surface **12b** of the SiC substrate **12**.

[0045] The SiC substrate **12** has a source region **30**, a contact region **32**, a body region **34**, a drift region **36**, and a drain region **38**. The source region **30** is an n-type region having a high n-type impurity concentration. The source region **30** is in contact with the source electrode **20** and the gate insulating film **14**. The contact region **32** is a p-type region having a high concentration of p-type impurities. The contact region **32** is in contact with the source electrode **20**. The body region **34** is a p-type region having a lower p-type impurity concentration than the contact region **32**. The body region **34** is in contact with the source region **30** and the contact region **32** from the lower side. The body region **34** is in contact with the gate insulating film **14** at a position below the source region **30**. The drift region **36** is an n-type region having a lower n-type impurity concentration than the source region **30**. The drift region **36** is in contact with the body region **34** from the lower side. The drift region **36** is in contact with the gate insulating film **14** at a position below the body region **34**. The drain region **38** is an n-type region having a higher n-type impurity concentration than the drift region **36**. The drain region **38** is in contact with the drift region **36** from the lower side. The drain region **38** is in contact with the drain electrode **22**.

[0046] When a potential equal to or higher than a threshold is applied to the gate electrode **16**, a channel is formed in the body region **34** along the gate insulating film **14** (along the side surface **24a** of the trench **24**). A channel connects the source region **30** and the drift region **36**. When a potential higher than that of the source electrode **20** is applied to the drain electrode **22** while the channel is formed, electrons flow from the source region **30** through the channel and the drift region **36** to the drain region **38**.

[0047] Next, a method for manufacturing the MOS transistor **10** will be described in the embodiment. According to the manufacturing method of the embodiment, it is possible to manufacture the MOS transistor **10** having a low on-resistance. The manufacturing method in the

embodiment includes forming a trench gate structure. In each drawing explaining the manufacturing method, illustration of the p-type region and the n-type region in the SiC substrate **12** is omitted.

[0048] As shown in FIG. 2A, the SiC substrate **12**, before the trenches are formed, is prepared. The SiC layer exposed at the upper surface **12a** is an epitaxial layer formed by epitaxial growth. Although not shown in FIG. 2A, the source region **30**, the contact region **32**, the body region **34**, the drift region **36**, and the drain region **38** are provided in the SiC substrate **12**.

[0049] As shown in FIG. 2B, the upper surface **12a** of the SiC substrate **12** is selectively etched to form the trench **24** in the upper surface **12a**.

[0050] A surface portion of the SiC substrate **12** has a high density of carbon defects having C—C bond. The carbon defects form interface states and trap electrons. If the carbon defects are present at a high density at the interface between the side surface **24a** of the trench **24** and the gate insulating film **14**, Coulomb scattering occurs due to electrons and the like captured at the interface states. In this case, the channel mobility decreases and the channel resistance increases. In contrast, as described below, in the manufacturing method of the embodiment, the gate structure can be formed in a state in which the density of interface states on the side surface **24a** of the trench **24** is low.

[0051] After forming the trench **24** as shown in FIG. 2B, high-temperature hydrogen etching is performed as shown in FIG. 3A. In the high-temperature hydrogen etching, the SiC substrate **12** is heated to a temperature of 1200° C. or higher (for example, 1300° C.) in hydrogen (H.sub.2) gas. Then, the surface of the SiC substrate **12** (specifically, the upper surface **12a** and the inner surface of the trench **24**) is etched by the hydrogen gas. Here, a very thin layer near the surface of the SiC substrate **12** is etched. As a result, carbon defects can be removed from the surface of the SiC substrate **12**. That is, the density of carbon defects on the surface of the SiC substrate **12** can be reduced by the high-temperature hydrogen etching.

[0052] Next, a step of forming an amorphous silicon layer shown in FIG. 3B is conducted. For example, a low-pressure CVD apparatus is used to form a silicon layer on the surface of the SiC substrate **12** in a nitrogen atmosphere. For example, SiH.sub.4 gas and N.sub.2 gas are used. The temperature for forming the film can be controlled to a temperature of 400° C. or more and 550° C. or less, and the pressure for forming the film can be controlled to a pressure of 0.3 Torr or more and 1.0 Torr or less. When CVD is performed under the low-temperature and low-pressure condition, the silicon layer grows at a slow speed. As a result, an amorphous silicon layer **42** is formed with a relatively uniform thickness on the upper surface **12a** of the SiC substrate **12** and on the inner surface of the trench **24**. FIG. 4 shows an electron diffraction image of the amorphous silicon layer **42** formed on the side surface of the trench **24** by this method. As shown in FIG. 4, a broad halo pattern is observed, and it is confirmed that the silicon layer formed on the side surface of the trench **24** is amorphous.

[0053] Furthermore, when CVD is performed under such conditions, the surface roughness of the amorphous silicon layer **42** can be reduced. For example, the flow rate of SiH.sub.4 gas can be controlled to 250 sccm, and the flow rate of N.sub.2 gas can be controlled to 230 sccm. The amorphous silicon layer **42** is formed so that the thickness of the amorphous silicon layer **42** on the side surface **24a** is 6 nm or less by adjusting the time for forming the film.

[0054] Next, a step of forming a gate insulating film shown in FIG. 3C is conducted. A silicon oxide layer **44** is formed on the surface of the amorphous silicon layer **42** using a low-pressure CVD apparatus. That is, the silicon oxide layer **44** is formed on the top of the upper surface **12a** and inside the trench **24**.

[0055] Next, nitrogen termination treatment is carried out. In the nitrogen termination treatment, the SiC substrate **12** is heated to a temperature of 1200° C. or higher (such as 1250° C.) in nitrogen gas (N.sub.2 gas) or nitrogen oxide gas (such as NO gas, N.sub.2O gas, N.sub.2 gas, and the like). This causes the SiC crystal to be nitrogen-terminated.

[0056] In a comparative manufacturing method, the surface of the SiC substrate is oxidized in the step of forming the gate insulating film. Furthermore, in a comparative manufacturing method, the surface of the SiC substrate may be oxidized during the nitrogen termination treatment. For example, when nitrogen oxide gas is used in the nitrogen termination treatment, the surface of the SiC substrate may be oxidized by oxygen atoms in the nitrogen oxide gas. Even when nitrogen gas is used in the nitrogen termination treatment, the surface of the SiC substrate may be oxidized by a small amount of oxidizing gas mixed into the chamber. When the surface of the SiC substrate is oxidized in the step of forming the gate insulating film and the nitrogen termination treatment, carbon defects are generated on the surface of the SiC substrate.

[0057] In contrast, in this embodiment, in the step of forming the gate insulating film and the nitrogen termination treatment shown in FIG. 3C, the amorphous silicon layer **42** is oxidized instead of the SiC substrate **12**. The silicon oxide layer **42a** formed by oxidizing the amorphous silicon layer **42** is integrated with the silicon oxide layer **44** formed by CVD. Thus, the gate insulating film **14** is formed. Furthermore, since the amorphous silicon layer **42** is oxidized instead of the SiC substrate **12**, oxidation of the surface of the SiC substrate **12** is suppressed. Therefore, carbon defects are unlikely to be generated on the surface of the SiC substrate **12** (specifically, the interface between the SiC substrate **12** and the gate insulating film **14**). As a result, the density of interface states on the surface of the SiC substrate **12** can be reduced. Due to this effect and the effect of terminating the SiC crystal with nitrogen by the nitrogen termination treatment, the density of interface states at the interface between the SiC substrate **12** and the gate insulating film **14** can be significantly reduced. In this embodiment, since the amorphous silicon layer **42** can be formed on the side surface **24a** of the trench **24**, the density of interface states can be significantly reduced on the side surface **24a** (specifically, the interface between the SiC substrate **12** and the gate insulating film **14**).

[0058] Next, as shown in FIG. 3D, the gate electrode **16** is formed in the trench **24**. In addition, the interlayer insulating film **18** is formed on the gate electrode **16**. This completes the trench gate structure.

[0059] Next, as shown in FIG. 1, the gate insulating film **14** on the upper surface **12a** is selectively removed to form a contact hole **20a**. Next, the source electrode **20** is formed to cover the upper surface **12a** of the SiC substrate **12**. Next, the drain electrode **22** is formed to cover the lower surface **12b** of the SiC substrate **12**. Through the above steps, the MOS transistor **10** shown in FIG. 1 is completed.

[0060] According to the manufacturing method of the embodiment, carbon defects present on the side surface **24a** of the trench **24** can be removed by high-temperature hydrogen etching. Furthermore, according to the manufacturing method of the embodiment, the amorphous silicon layer **42** is oxidized in the step of forming the gate insulating film and the nitrogen termination treatment, so that oxidation of the SiC substrate **12** can be suppressed and the generation of carbon defects on the side surface **24a** of the trench **24** can be suppressed. Moreover, according to the manufacturing method of the embodiment, the density of interfaces state at the interface between the SiC substrate **12** and the gate insulating film **14** can be reduced by the nitrogen termination treatment. Therefore, it is possible to manufacture the MOS transistor **10** having a low density of interface states on the side surface **24a** (specifically, the interface between the body region **34** and the gate insulating film **14**). Therefore, according to this manufacturing method, the channel resistance of the MOS transistor **10** can be reduced. That is, according to this manufacturing method, the MOS transistor **10** having a low on-resistance can be manufactured.

[0061] The characteristics of the MOS transistor **10** manufactured by the manufacturing method of the embodiment (hereinafter referred to as the embodiment prototype) and a MOS transistor manufactured by a conventional manufacturing method (hereinafter referred to as the conventional product) are evaluated. It is confirmed that the embodiment prototype could achieve a higher channel mobility than the conventional product. Also, in the conventional product, the channel

mobility decreases as the temperature decreases, whereas in the embodiment prototype, the channel mobility increases as the temperature decreases. This is believed to be because in the embodiment prototype, the density of interface states is reduced, resulting in a reduction in the influence of Coulomb scattering at low temperature. Moreover, it is confirmed that the embodiment prototype has a greater effect of reducing the on-resistance of the element than the conventional product. This is believed to be because in the embodiment prototype, the channel resistance is reduced due to the increased channel mobility. It is also confirmed that the embodiment prototype has the same withstand voltage as the conventional product.

[0062] FIGS. 5 to 7 show cross sections, at the upper surface **12a**, along a direction perpendicular to an extending direction of the trench **24** and along a thickness direction of the SiC substrate **12**. FIGS. 5 and 6 show enlarged views of the boundary portion (hereinafter referred to as shoulder portion) between the side surface **24a** and the upper surface **12a** when the gate insulating film **14** is formed by the manufacturing method of the embodiment. More specifically, FIG. 5 shows the shape of the shoulder portion when the sacrificial oxide film forming step and the sacrificial oxide film removing step are not performed, and high-temperature hydrogen etching is performed for 15 minutes. FIG. 5 corresponds to the MOS transistor **10** manufactured by the manufacturing method of the embodiment (hereinafter, referred to as the embodiment prototype). FIG. 6 shows the shape of the shoulder portion when the sacrificial oxide film forming step and the sacrificial oxide film removing step are not performed and high-temperature hydrogen etching is performed for 30 minutes. FIG. 7 shows an enlarged view of a shoulder portion when the gate insulating film **14** is formed by the manufacturing method of the comparative example. More specifically, FIG. 7 shows the shape of the shoulder portion when the sacrificial oxide film forming step and the sacrificial oxide film removing step are performed, and argon etching is performed instead of high-temperature hydrogen etching.

[0063] In argon etching, the etching rate is significantly different between the upper surface **12a** and the side surface **24a**, and etching of the upper surface **12a** proceeds faster than etching of the side surface **24a**. For this reason, as shown in FIG. 7, an inclined portion **100** is formed in the shoulder portion, which is gradually sloped downward as approaching the trench. Furthermore, in a bent portion **102** between the inclined portion **100** and the side surface **24a**, the surface of the SiC substrate **12** is curved with a small radius of curvature. That is, in FIG. 7, the radius of curvature of the shoulder portion is the smallest at the bent portion **102**. Specifically, the value of the radius of curvature is 47 nm at the bent portion **102**. A dashed line **200** shown in FIG. 7 indicates a shoulder portion connecting the upper surface **12a** and the side surface **24a** as an elliptical arc. The aspect ratio of the ellipse of the elliptical arc **200** is 0.40. The aspect ratio (flattening ratio) f is a value expressed by $f=1-b/a$, where “ a ” is the semimajor axis of an ellipse and “ b ” is the semiminor axis of the ellipse. In the argon etching, the etching rate of the upper surface **12a** is faster than that of the side surface **24a**, so that the aspect ratio of the elliptical arc **200** is large.

[0064] In contrast, according to the high-temperature hydrogen etching, the upper surface **12a** and the side surfaces **24a** are etched more uniformly than argon etching. Therefore, as shown in FIGS. 5 and 6, a locally small radius of curvature (such as the bent portion **102** in FIG. 7) is not formed in the shoulder portion between the upper surface **12a** and the side surface **24a**. Therefore, the high-temperature hydrogen etching increases the minimum radius of curvature at the shoulder portion. For example, the minimum radius of curvature of the shoulder portion is 53 nm in FIG. 5, and is 71 nm in FIG. 6. A dashed line **202**, **204** in FIGS. 5 and 6 show a shoulder portion connecting the upper surface **12a** and the side surface **24a** as elliptical arc. In FIG. 5, the aspect ratio of the elliptical arc **202** is 0.29. In FIG. 6, the aspect ratio of the elliptical arc **204** is 0.25. That is, the aspect ratio of the elliptical arc in FIGS. 5 and 6 is smaller than that in FIG. 7. That is, the elliptical arc in FIGS. 5 and 6 is closer to a perfect circle arc than that in FIG. 7. In the high-temperature hydrogen etching, the upper surface **12a** and the side surface **24a** are etched more uniformly than in the argon etching, so that the aspect ratio of the elliptical arc **202**, **204** can be made smaller.

[0065] FIG. 8 shows the results of evaluating the breakdown voltage of the gate insulating film 14 at the shoulder position. The horizontal axis of FIG. 8 represents the voltage applied to the gate insulating film 14, and the vertical axis of FIG. 8 represents the leakage current flowing through the gate insulating film 14. Graphs G5, G6, and G7 in FIG. 8 respectively show the results for the MOS transistors of FIGS. 5, 6 and 7. As shown in FIG. 8, the graphs G5 and G6 provide higher withstand voltage than the graph G7. It can be seen from FIG. 8 that when the minimum value of the radius of curvature of the shoulder portion is 53 nm or more and the aspect ratio of the elliptical arc of the shoulder portion is 0.29 or less, the withstand voltage is stable at a high value. In this way, the high-temperature hydrogen etching can form the shoulder portion into a smoother curved shape, making it possible to increase the withstand voltage.

[0066] In the embodiment, a method for manufacturing a trench MOS transistor has been described. However, the techniques disclosed herein may also be applied to a manufacturing method of MOS transistor having a planar gate structure. In this case, the technique disclosed in this specification makes it possible to form a gate insulating film to cover the main surface (specifically, the upper surface) of the SiC substrate.

[0067] Although the embodiment has been described in detail above, these are merely examples and do not limit the scope of present disclosure. The techniques described in claims include various modifications of the specific examples illustrated above. The technical elements described in the present specification or the drawings exhibit technical usefulness alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. In addition, the techniques illustrated in the present specification or drawings achieve plural objectives at the same time, and achieving one of the objectives itself has technical usefulness.

Claims

1. A method of manufacturing a MOS transistor comprising: etching a surface of a SiC substrate with hydrogen gas in a state where the SiC substrate is heated to a temperature of 1200° C. or higher; forming an amorphous silicon layer on the surface of the SiC substrate after the etching; forming a gate insulating film made of silicon oxide on a surface of the amorphous silicon layer by chemical vapor deposition; and performing a nitrogen termination treatment on the SiC substrate after the forming of the gate insulating film.
2. The method according to claim 1, wherein a trench is provided in a main surface of the SiC substrate, the etching includes etching a side surface of the trench with hydrogen gas, the forming of the amorphous silicon layer includes forming the amorphous silicon layer on the side surface of the trench, and the forming of the gate insulating film includes forming the gate insulating film on the surface of the amorphous silicon layer covering the side surface of the trench.
3. The method according to claim 2, wherein the etching includes etching the side surface of the trench and the main surface of the SiC substrate, the forming of the amorphous silicon layer includes forming the amorphous silicon layer on the side surface of the trench and on the main surface of the SiC substrate, and the forming of the gate insulating film includes forming the gate insulating film on the surface of the amorphous silicon layer within an area covering the side surface of the trench and the main surface of the SiC substrate.
4. The method according to claim 1, wherein the amorphous silicon layer is formed by chemical vapor deposition.
5. The method according to claim 4, wherein the amorphous silicon layer is formed in a nitrogen atmosphere by chemical vapor deposition in a state where the SiC substrate is heated to a temperature of 400° C. or more and 550° C. or less.
6. The method according to claim 4, wherein the amorphous silicon layer is formed by chemical vapor deposition in a state where the SiC substrate is placed under a pressure of 0.3 Torr or more and 1.0 Torr or less.

7. The method according to claim 1, wherein the amorphous silicon layer has a thickness of 6 nm or less.

8. The method according to claim 1, wherein a main surface of the SiC substrate is a surface of an epitaxial layer.

9. The method according to claim 1, wherein the nitrogen termination treatment includes heating the SiC substrate to a temperature of 1200° C. or higher in nitrogen gas or nitrogen oxide gas.

10. The method according to claim 2, wherein the main surface of the SiC substrate and the side surface of the trench are connected by an arc having a minimum radius of curvature of 53 nm or more, in a cross section of the main surface of the SiC substrate along a direction perpendicular to the trench and a thickness direction of the SiC substrate, after the gate insulating film is formed.

11. The method according to claim 2, wherein the main surface of the SiC substrate and the side surface of the trench are connected by an elliptical or circular arc having an aspect ratio of 0.29 or less, in a cross section of the main surface of the SiC substrate along a direction perpendicular to the trench and a thickness direction of the SiC substrate, after the gate insulating film is formed.

12. A trench gate MOS transistor comprising: a SiC substrate having a trench provided on a main surface of the SiC substrate; a gate insulating film covering a side surface of the trench; and a gate electrode provided in the trench, wherein the main surface of the SiC substrate and the side surface of the trench are connected by an arc having a minimum radius of curvature of 53 nm or more in a cross section of the main surface along a direction perpendicular to the trench and a thickness direction of the SiC substrate.

13. A trench gate MOS transistor comprising: a SiC substrate having a trench provided on a main surface of the SiC substrate; a gate insulating film covering a side surface of the trench; and a gate electrode provided in the trench, wherein the main surface of the SiC substrate and the side surface of the trench are connected by an elliptical or circular arc having an aspect ratio of 0.29 or less in a cross section of the main surface along a direction perpendicular to the trench and a thickness direction of the SiC substrate.
